1.What are the four pillars of OOP?

The four pillars are: encapsulation, abstraction, inheritance, and polymorphism. Encapsulation is the organisation of data and methods within one unit (a class) and the limitation of direct access to some of its elements.

Abstraction hides complex implementation details and shows only what is necessary. The process of inheriting properties and methods from a parent class to a child's class is called inheritance.

Polymorphism is the ability of a single method to react in separate ways to different objects that call it.

2.What is the difference between method overloading and method overriding?

When a method with the same name exists more than once in the same class, each with a separate set of parameters, it is called method overloading (compile-time polymorphism).

The method of overriding is the process of defining a method in a subclass that already exists in the parent class, which is known as runtime polymorphism.

When a method is already defined in the parent class, but then redefined in the subclass, this is known as runtime polymorphism, or method overriding. Overloading is resolved at compile time and overriding at run time.

3. Can you explain encapsulation with an example?

Encapsulation is the process of bundling data (variables) and code (methods) into one unit and restricting access to the code using access modifiers.

Private variables having public getters and setters are examples in Java. This helps prevent direct modification of the internal state by external code.

4. What is an abstract class, and how does it differ from an interface?

An abstract class can have both abstract methods (without implementation) and concrete methods (with implementation). Traditional Java (pre-Java 8) has a concept of an interface: it has just abstract methods.

From Java 8 onwards, interfaces can have default and static methods. A class may extend any number of abstract classes but implement only a few interfaces.

5. What is the diamond problem in OOP?

The diamond problem is a multiple inheritance issue where a class is derived from two classes which are both derived from a common base class. This leads to confusion over which parent's method should be invoked.

6. Explain normalisation up to 3NF.

Normalisation is the process of organising a database to reduce redundancy.

- First Normal Form (1NF) requires that each column contains only atomic (indivisible) values.
- Second Normal Form (2NF) requires 1NF plus removal of partial dependencies every non-key column must depend on the entire primary key.
- Third Normal Form (3NF) requires 2NF plus removal of transitive dependencies non-key columns must not depend on other ones.

Normal Form	Rule	Example of Violation
1NF	Atomic values only	A "Phone" column storing multiple numbers separated by commas

2NF	No partial dependency	A column depending on only part of a composite primary key
3NF	No transitive dependency	“City” depending on “Zip Code” instead of the primary key

7. What is the difference between a primary key and a foreign key?

A primary key uniquely identifies each record in a table and cannot be null. A foreign key is a column (or set of columns) in one table that references the primary key of another table.

It creates a relationship between two tables and can accept null values unless explicitly constrained.

8. What is a deadlock in DBMS?

Deadlocks happen when two or more transactions are waiting on one another to unlock other transactions, creating a “wait cycle.” Neither can proceed.

Deadlocks are prevented, detected, and resolved in DBMS systems by ordering lock requests, wait-for graphs, and timeout.

9. What is the difference between WHERE and HAVING?

WHERE and HAVING both do the same thing; WHERE is used in the FROM clause and HAVING is used in the WHERE clause.

WHERE filters the rows prior to grouping (used with individual rows). WHERE used after GROUP BY (when using aggregate functions such as COUNT, SUM, AVG).

10. Write a query to find the second-highest salary from an employee table.

SELECT MAX(salary) AS SecondHighest

FROM Employee

WHERE salary < (SELECT MAX ((salary) FROM Employee);

11. Explain several types of JOINs with examples.

INNER JOIN will only return matching rows from both tables. LEFT JOIN fetches all rows from the left table and the right table if there is a match (no rows if there is no match) The RIGHT JOIN does the opposite.

FULL OUTER JOIN returns all the rows from both tables, including any rows that have no matching rows. CROSS JOIN will return the Cartesian product of both tables.

12. What is the subquery?

A subquery is a query that is contained within a query. It may be used in the WHERE clause or the FROM clause, or the SELECT clause. Subqueries can be used when the filtering condition is based on a different calculation.

13. What is the difference between a stack and a queue?

A stack is a Last in First Out data structure — that is, the last element added to the stack is the first element removed from the stack.

A queue is a data structure that follows First in First Out (FIFO) — the first element that is inserted into a queue is the first one to be taken out.

Stacks can be used in undo operations and the function of call management. In scheduling and breadth-first search, queues are used.

14. Explain the telicity of common sorting algorithms.

Algorithm	**Best Case**	**Average Case**	**Worst Case**	**Stable?**
Bubble Sort	O(n)	O(n²)	O(n²)	Yes

Merge Sort	O ((n log n)	O ((n log n)	O ((n log n)	Yes
Quick Sort	O ((n log n)	O ((n log n)	$O(n^2)$	No
Insertion Sort	O(n)	$O(n^2)$	$O(n^2)$	Yes

15. What is a linked list, and how does it differ from an array?

A link list is a data structure that is like an array but with different data elements represented in the form of nodes. In a Linked List, the elements are stored as nodes;nodes;node has data along with a pointer to the next node.

The elements in an array are stored in contiguous memory. Random access is O(n) in a linked list while insertion and deletion are efficient at known positions (O (1)). Arrays: Random access:O ((1) Insertion/Deletion: O(n)

16. What is binary search, and when can it be used?

Binary search is a divide and conquer search algorithm that only works when the data is sorted. Binary search is applicable to sorted arrays.

It continually halves the interval to search and compares the middle element with the target. The time complexity of this algorithm is O (log n). It cannot be applied to unsorted data without first sorting it.

17. What is the difference between C and C++?

C is a procedural programming language. C++ is a procedural, and an object-oriented language. C++ supports features not available in C such as classes, inheritance, polymorphism, encapsulation, exception handling and so on.

18. Give one key difference about the list and a tuple in Python?

Lists can be modified after creation; Tuples cannot be. Lists are represented by square brackets []; Tuples are represented by parentheses (). The immutability of tuples makes them slightly faster and use less memory.

19. Explain the meaning of Python decorators?

Decorators are functions that alter the behavior of other functions without altering the source code. They are used in the following way: above a function definition. This is typically used for logging, access control, and caching.

20. How does exception handling work in Python?

Exception handling in Python is managed using try, except, else, and finally blocks. Code that may cause an error is placed inside the try block. If an exception occurs, the except block handles the error and prevents the program from terminating abruptly.

The other block executes only when no exception occurs in the try block. Thefinalblock runs regardless of whether an exception is raised, making it useful for cleanup operations such as closing files or releasing resources.